

ST FERGUS GAS TERMINAL

Design and cybersecurity of SCADA application



Px group manages the St Fergus Gas Terminal in Scotland. Process Safety Solutions (PSS), a partner system integrator of Panorama, aids in maintaining and upgrading Panorama SCADA applications at the site. Px group plans to update the software to adhere to Oil & Gas and cybersecurity standards due to installation criticality. To achieve this, PSS has fully upgraded the HMI Panorama for emergency shutdown (ESD), fire and gas (F&G), and burner management systems (BMS). The upgraded system was commissioned without any plant downtime.



Aim

The HMI upgrade's goal is EEMUA 201 and IEC 62443 alignment. EEMUA 201 supplements IEC 62443, offering precise critical system security insights. The Panorama software solution has been installed at this key UK onshore gas terminal for several years. **Backward compatibility** and **scalability** of the software allow the customer's investments in the SCADA application and its environment to be sustainable by making them scalable and maintainable over time.



Solution

Panorama's solution allows the integrator to implement three application guidelines, which are:

- **Design of the application:** limited use of colour and grey backgrounds to minimize glare and low-contrast depictions in 2D, not 3D.
- **Alarm management:** no animation except for alarm management and, analogue representation of important measurements, indicating their value (normal, abnormal, and alarm conditions).
- **Display hierarchies and navigation path:** a proper hierarchy of display content providing for the progressive exposure of detailed information as needed, logical and consistent navigation methods and consistent process flow depiction and layout to minimise crossing lines.

New redundant servers and operator stations were deployed with the latest operating systems and all servers and PCs have been **cybersecurity hardened**. All HMI operator screens, static elements (vessels, heat exchangers etc.), and dynamic elements (valves, analogue indicators, histograms etc.) were also updated.

Customer benefits

The use of EEMUA 201 and IEC 62443 guidelines allows for **improved operator performance, plant efficiencies and safety**. Reviews were completed with Control Room Operators on the design and new functionality. The user experience is thus considered in all developments, to respond effectively to requests from the field and make the application as user-friendly as possible.

